

Simplicity as a Feature: Robust Transition-State Search by Gentlest Ascent Dynamics with Cheap Analytic Hessians

June 25, 2026

Abstract

Machine-learned interatomic potentials (MLIPs) make analytic Hessians cheap (≈ 0.06 s/structure with HIP [9]), so the bottleneck in transition-state (TS) search is no longer the Hessian but the *step*. We use the minimal choice — first-order Gentlest Ascent Dynamics (GAD) on the lowest eigenvector of the Eckart-projected analytic Hessian, a Cartesian step with no trust region, quasi-Newton update, RFO, or eigenvector prediction. Our primary metric is the IRC-validated *intended success rate* (forward and backward IRC must graph-match the intended reactant and product), on 287 Transition1x test reactions from perturbed-saddle guesses (10–200 pm).

Highlights

- **Crossover, not a point.** GAD is statistically indistinguishable from the strongest Sella baseline when the guess is good (McNemar $p \approx 1.0$, ≤ 50 pm) and pulls significantly ahead as it degrades ($p < 10^{-4}$ by 150–200 pm; **+21.6 pp intended at 200 pm**, 45.3% vs. 23.7%). The same crossover holds for convergence and across $F_{\max}$ thresholds.
- **Mechanism (corrected).** No method converges to an unintended (wrong-basin) saddle — unintended $\approx 0/287$ for all three. GAD’s edge is recovering the full two-sided reaction more often; Sella’s high-noise loss is non-convergence + one-sided (partial) saddles, not wrong basins.
- **Honest cost.** First-order GAD cannot tighten $F_{\max}$ below ≈ 0.01 eV/Å; no step budget removes it. This plateau lives on the *secondary* (convergence) axis only.
- **Hybrid.** A GAD→Newton switch removes the plateau, wins wall-time per converged TS (1.4–3 $\times$) and geometry (tightest p95 RMSD), and inherits GAD’s chemistry.
- **Scope boundary.** Robustness is demonstrated for *perturbed-saddle* guesses. From a reactant start the GAD family is weak (hybrid $\sim 2\%$) and Sella wins — GAD needs to start near the saddle for the reaction mode to be identifiable.

1 Introduction

- TSs are index-1 saddle points; locating them needs curvature (the Hessian, or at least its leftmost eigenpair). *Ab initio* Hessians are expensive, so optimizers avoid them: dimer [2], eigenvector-following / P-RFO [3, 4, 5], Sella’s iterative partial-Hessian diagonalization [6, 7], and recent ML prediction of the Hessian [10, 9] or its leftmost eigenvector [11].
- **The premise has changed.** With MLIPs the analytic Hessian is cheap (≈ 0.06 s for HIP [9]). The bottleneck becomes the **step**. Trust-region/Newton jumps ($\mathcal{O}(0.1$ Å)) are optimal near a quadratic saddle but fragile from a poor guess.

- **This work.** Given a cheap exact Hessian, discard P-RFO, trust radii, QN, and eigenvector prediction; step with first-order GAD [1]. Score success by IRC-validated *intended* recovery [10, 11], not mere convergence. Thesis: *in the MLIP regime you need not avoid the Hessian — you need to step more carefully with it.*

2 Methods

GAD dynamics. With forces $\mathbf{F} = -\nabla E$ and $\mathbf{v}_1$ the lowest eigenvector of the mass-weighted, Eckart-projected Hessian,

$$\dot{\mathbf{x}} = (\mathbf{I} - 2\mathbf{v}_1\mathbf{v}_1^\top)\mathbf{F} = \mathbf{F} - 2(\mathbf{F}\cdot\mathbf{v}_1)\mathbf{v}_1, \quad \mathbf{x}_{k+1} = \mathbf{x}_k + dt\dot{\mathbf{x}}. \quad (1)$$

- **Step:** fixed- dt explicit Euler ($dt \in \{0.003, 0.005, 0.007\}$; representative 0.005), Cartesian, no history/trust-region/line-search; mode tracked by max overlap; per-atom cap 0.35 Å.
- **Potential:** HIP [9], an $SE(3)$ -equivariant MLIP returning the analytic Hessian (≈ 0.06 s); not trained on Transition1x (held out); Hessian recomputed every step.
- **Eckart projection:** remove the 6 translation/rotation modes in mass-weighted coordinates, diagonalize the reduced $(3N-6)$ Hessian; single most important robustness ingredient. (Cartesian vs. MW step: ≤ 0.35 pp.)
- **Hybrid GAD–Newton:** walk with Eq. (1); near a clean index-1 saddle ($n_{\text{neg}}=1$, small force) switch to a trust-radius-capped eigenvector-following Newton step on the same analytic Hessian.
- **Benchmark:** 287 Transition1x [8] test reactions; start = true saddle + isotropic Gaussian noise $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm; baseline = Sella [7] (strongest config: Cartesian + Eckart, exact Hessian/step; Appendix D).

Metrics — primary vs. secondary.

- **Convergence rate (secondary):** optimizer reached a first-order saddle, $n_{\text{neg}}=1 \wedge F_{\text{max}} < \tau$ (default $\tau = 0.01$ eV/Å; threshold family in Appendix C). Necessary but *insufficient*.
- **Intended success rate (primary):** after optimization, follow Sella’s IRC [7] forward/backward and graph-match each endpoint (Open Babel connectivity + VF2) to the labelled reactant/product. Outcomes (after [10, 11]): *intended* (both endpoints), *partially intended* (one), *unintended* (valid saddle, other reaction), *no reaction* (collapsed to a minimum), *TS error* (no convergence).

3 Results

3.1 Primary: intended success rate crosses over with guess quality

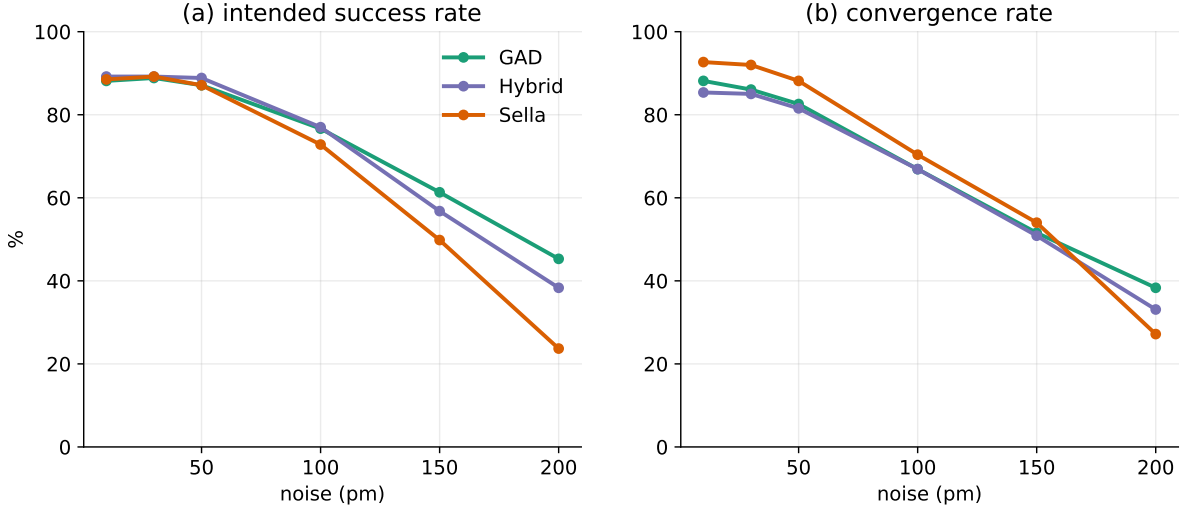


Figure 1: Primary (a) and secondary (b) axes vs. noise, $n = 287$. Methods coincide at low noise; GAD and the hybrid retain far more intended reactions as the guess degrades.

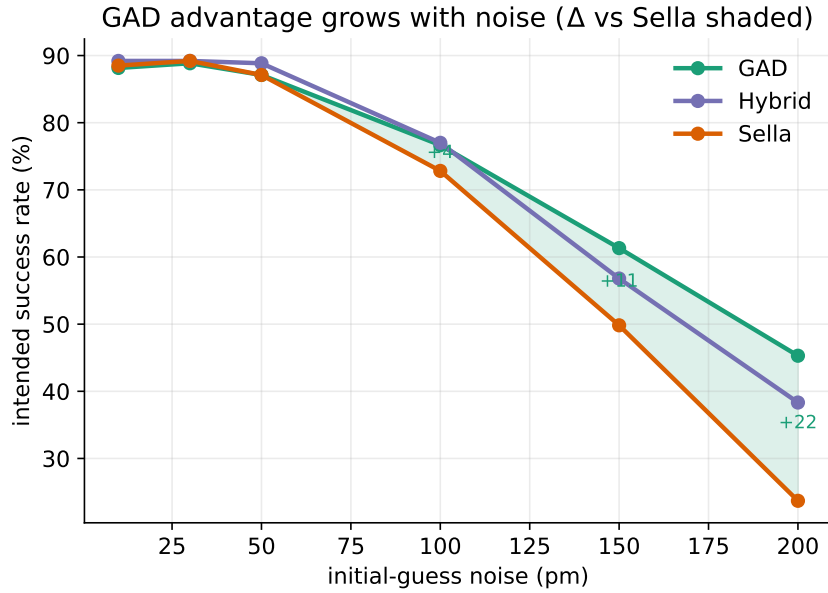


Figure 2: Intended success for all three methods; the GAD–Sella gap is shaded. Paired-McNemar (GAD vs. Sella): indistinguishable ≤ 50 pm ($p=1.0$), emerging at 100 pm ($p=0.10$), highly significant at 150 pm ($p=3.8 \times 10^{-5}$) and 200 pm ($p=1.7 \times 10^{-13}$).

- **Crossover.** Tie at low noise (intended 88–89% at 10 pm; Sella even leads on raw *convergence*, 92.7 vs. 89.2%); GAD pulls ahead with degradation: $\Delta_{\text{intended}} = +3.9, +11.5, +21.6$ pp at 100/150/200 pm (Fig. 2, Table 1).
- **Significance.** McNemar on paired per-sample intended outcomes: n.s. at ≤ 50 pm ($p=1.0$), emerg-

ing at 100 pm ($p=0.10$), highly significant at 150 pm ($p=3.8 \times 10^{-5}$) and 200 pm ($p=1.7 \times 10^{-13}$). Same crossover on convergence and across $F_{\max}$ thresholds.

Table 1: Intended success rate (primary) and convergence rate (secondary), %, $F_{\max} < 0.01$, $n = 287$. Intended from the fresh per-sample IRC re-validation (GAD $dt=0.003$); Sella = strongest baseline.

		10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
Intended %	GAD (ours)	88.2	88.9	87.1	76.7	61.3	45.3
	Sella (baseline)	88.5	89.2	87.1	72.8	49.8	23.7
	Δ (pp)	-0.3	-0.3	0.0	+3.9	+11.5	+21.6
Conv. %	GAD (ours)	89.2	88.5	85.4	71.1	55.1	40.8
	Sella (baseline)	92.7	92.0	88.2	70.7	54.0	27.2

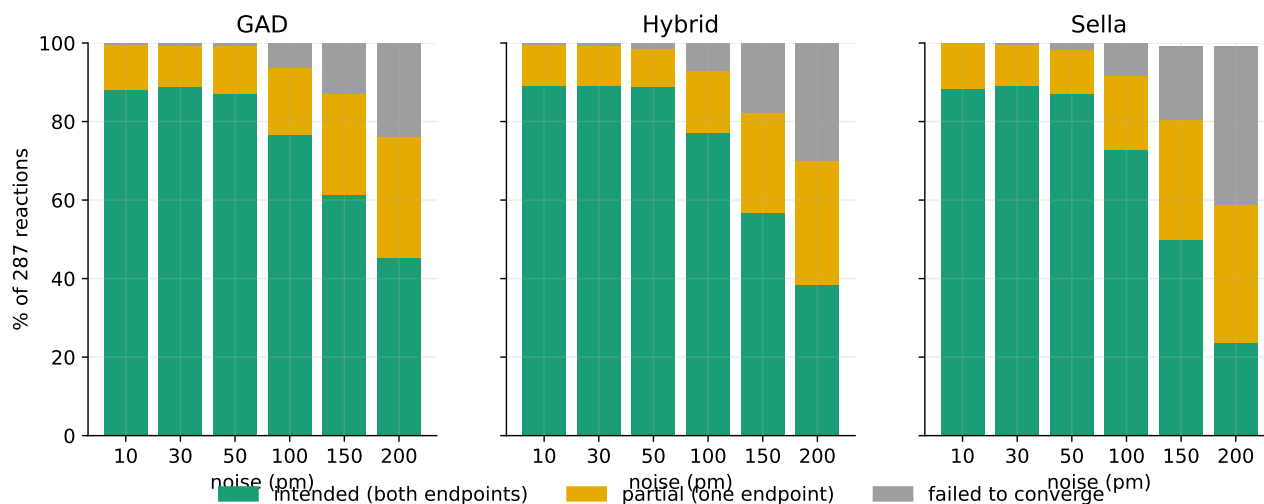


Figure 3: Full IRC outcome breakdown. $Unintended \approx 0$ for every method at every noise, so the third band is purely non-convergence (TS error / no reaction). Sella both fails to converge *and* lands one-sided (partial) saddles more often at high noise.

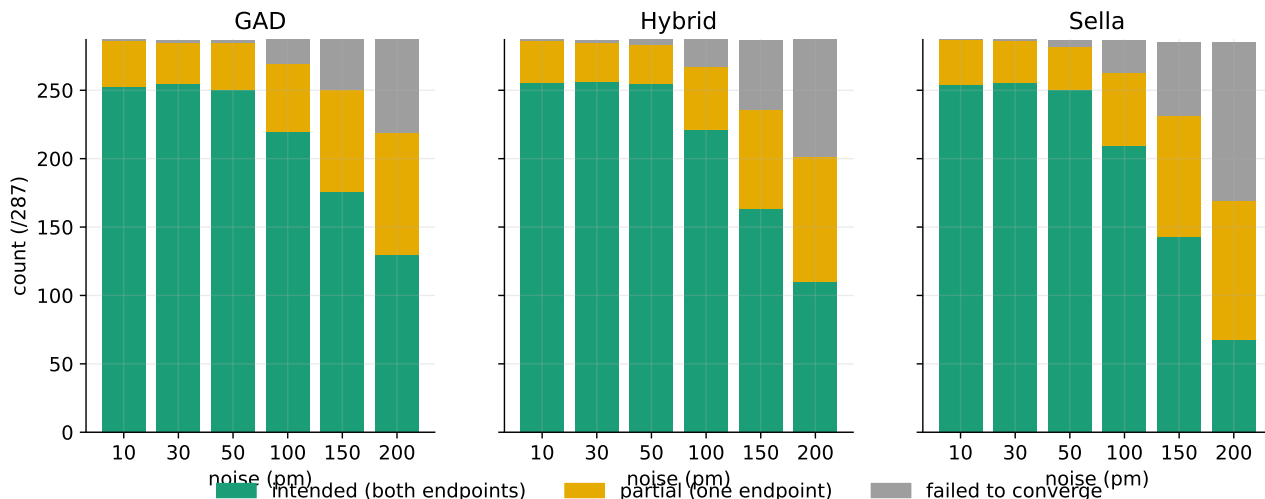


Figure 4: Same breakdown, absolute counts (/287). At 200 pm: intended/partial/failed = GAD 130/89/68 vs. Sella 68/101/116 — Sella’s loss is non-convergence (116 vs. 68) plus one-sided partial saddles (101 vs. 89).

3.2 Mechanism: it is not wrong basins — it is one-sided saddles and non-convergence

- **Unintended ≈ 0 .** The per-sample 5-tier IRC split gives unintended ≈ 0 for every method at every noise: 0 for GAD and the hybrid, and only 2 (150 pm) + 2 (200 pm) for Sella — **4 of 5166** samples total. *We therefore drop the earlier “Newton jumps across basin boundaries to unintended saddles” claim — it is unsupported by the data.*
- **What actually differs** (200 pm illustration; whole curves in Fig. 3): intended/partial/failed = **GAD 130/89/68** vs. **Sella 68/101/116**. Sella both fails to converge more *and* lands one-sided (partial) saddles more.
- **Why.** Small in-basin Euler steps preserve the link to *both* reactant and product, so even non-converged GAD geometries IRC-validate on both sides — intended can *exceed* convergence (150 pm: 61.3 vs. 55.1%; 200 pm: 45.3 vs. 40.8%). Larger Newton steps more often drift off one channel ($\rightarrow$ partial) or fail to tighten ($\rightarrow$ failed).
- **Convergence $\neq$ chemistry, stated correctly** (Fig. 5): it is *not* that convergence hides wrong saddles. (i) GAD recovers intended even when not strictly converged; (ii) Sella d=1 vs. d=3 — d=3 *converges more* but is *intended less* at every noise (a stale Hessian yields more partial/failed, not unintended).
- **Non-converged GAD breakdown** (why it fails to *converge*, not why it finds a wrong saddle): at 200 pm, of 163 non-converged, 71 (44%) drift to $n_{\text{neg}} \geq 2$ and 92 (56%) are plateau-orbits at $n_{\text{neg}}=1$ just above threshold (Section 3.3).

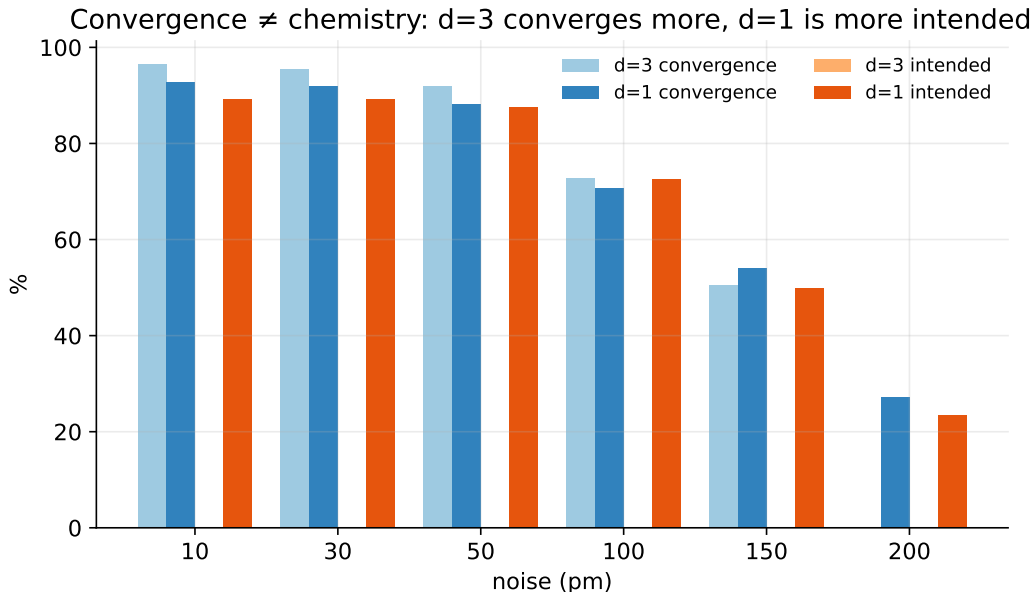


Figure 5: Convergence $\neq$ chemistry. Sella Hess.Freq.=3 converges more than Hess.Freq.=1 at low noise (a) yet recovers fewer intended reactions at every noise (b): the extra “convergences” are one-sided (partial) saddles, not unintended ones.

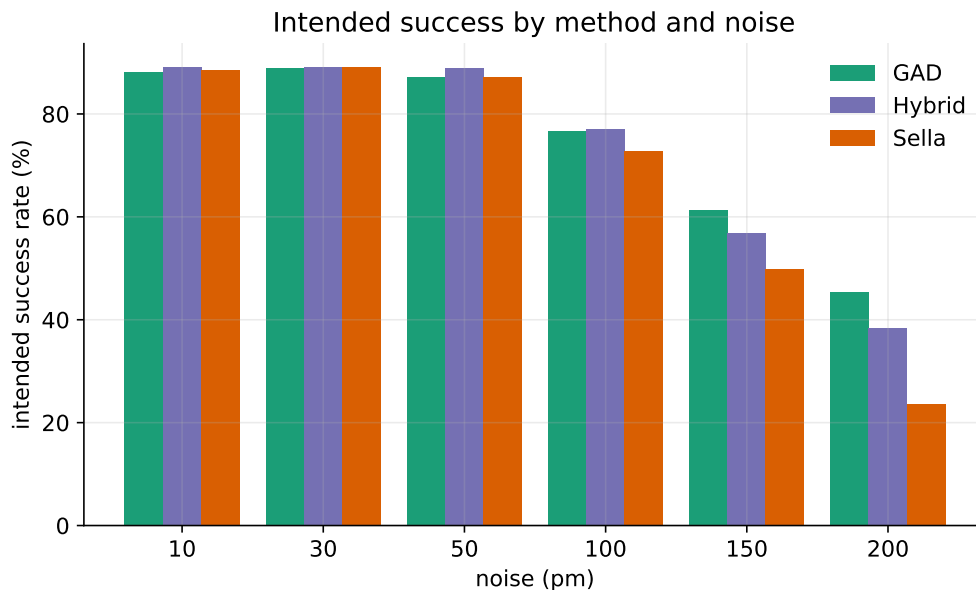


Figure 6: Intended reactions recovered (counts/287). The GAD family overtakes the baseline from 100 pm onward.

3.3 Secondary: the $F_{\max}$ plateau (and why it does not matter for chemistry)

- **Plateau.** All GAD dt collapse to 0% at $F_{\max} < 0.005$ at every noise; Sella’s Newton step retains 24–34% (low noise). Pure first-order dynamics has no samples in 0.001–0.005 eV/Å (Figs. 7, 8).
- **Mechanism.** GAD step $\sim dt \|\mathbf{F}\|$ shrinks linearly as $\mathbf{F} \rightarrow 0$ and falls below the eigenvector noise floor; Newton $\sim \|\mathbf{H}^{-1}\mathbf{F}\|$ stays finite. Structural to fixed-step first-order dynamics; larger dt trades

it for instability.

- **Intrinsic.** $5\times$ budget (10k vs. 2k steps, 50 pm) leaves $\tau=0.01$ unchanged (85.7%) and $\tau<0.005$ exactly 0.0%.
- **It lives on the secondary axis.** Intended success (Section 3.1) needs no sub-plateau force, so the plateau does not affect the primary metric.

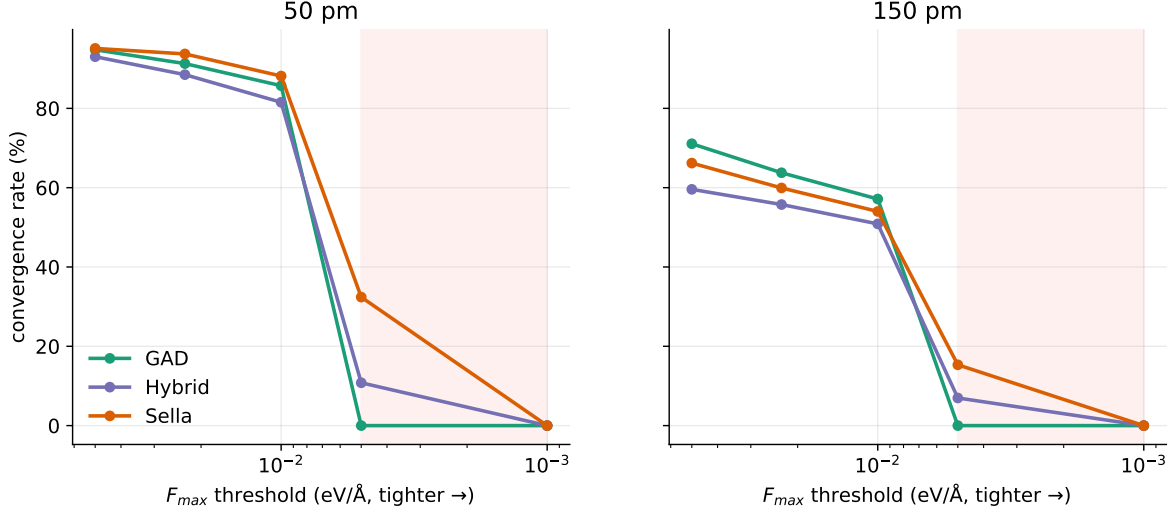


Figure 7: The $F_{\max}$ plateau (convergence vs. threshold; tighter to the right). GAD drops vertically near 0.01; Sella and the hybrid descend into 0.005 via Newton landing.

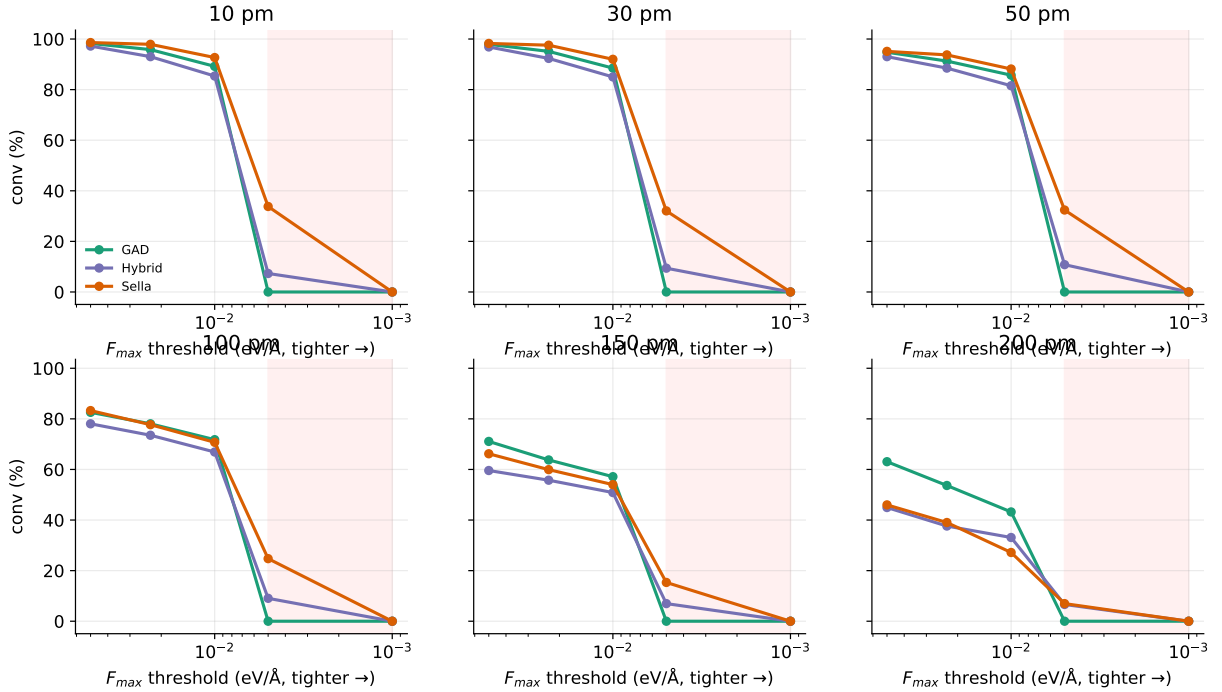


Figure 8: Convergence vs. $F_{\max}$ threshold, all six noise levels. The GAD cliff at 0.005 is universal.

3.4 Cost and geometry: the hybrid wins both

- **Wall-time / converged TS:** hybrid lowest at every noise — $1.3\text{--}3.0\times$ vs. Sella, $3.6\text{--}4.5\times$ vs. GAD (Fig. 9, Table 2). Few-but-expensive Sella steps (4–13) vs. many-cheap GAD steps (100–738) vs. intermediate-cheap hybrid (6–95) (Fig. 9). Pareto-dominant at high noise (Fig. 10).
- **Geometry:** converged-TS RMSD-to-reference — hybrid holds the tightest p95 at high noise (200 pm: **0.109** vs. GAD 0.456 vs. Sella 0.838 Å; $7.7\times$ tighter than Sella). Sella’s tail is bimodal (Fig. 10, Table 2).

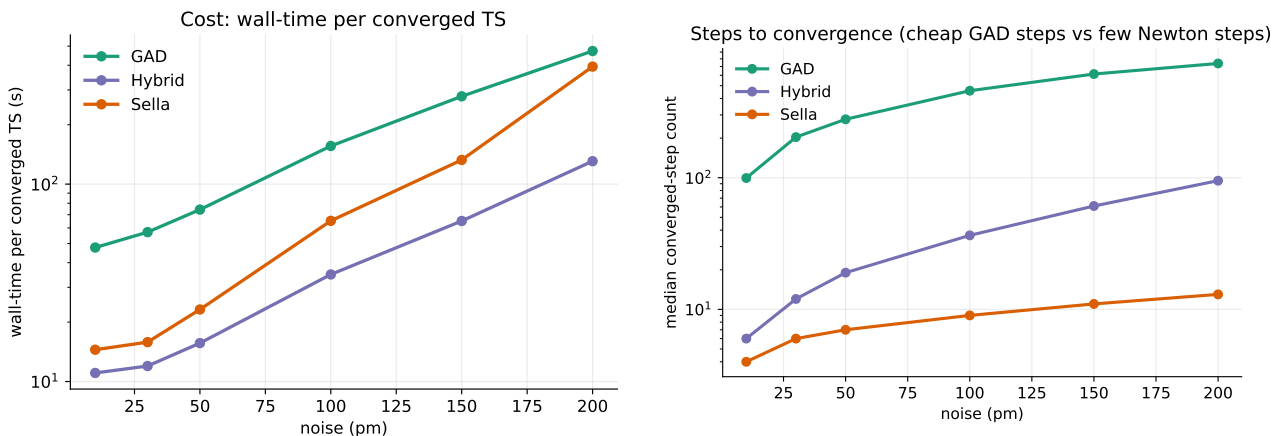


Figure 9: Cost. Wall-time per converged TS (left) and median steps (right). The hybrid is fastest despite more steps than Sella, because its steps are cheap.

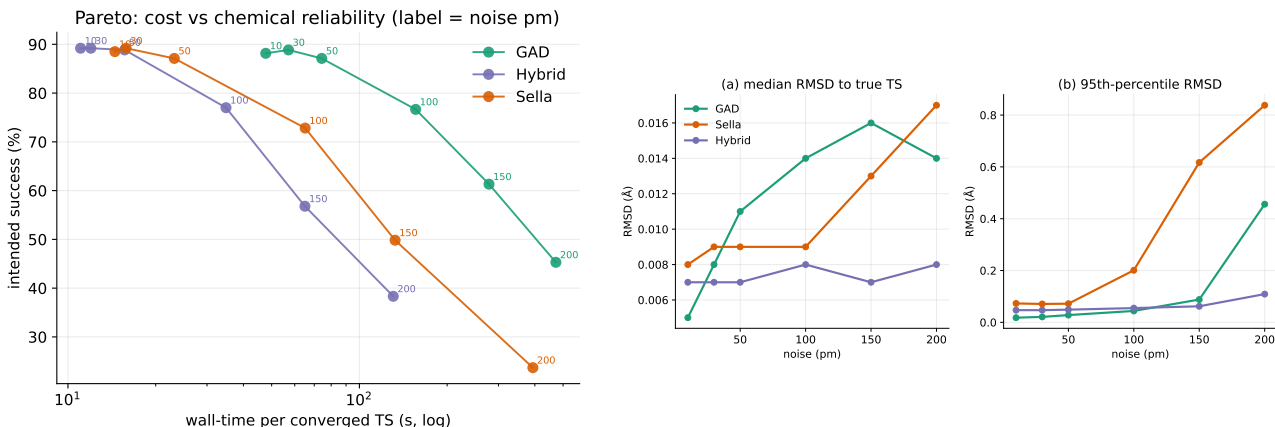


Figure 10: Left: Pareto (intended vs. wall-time; up-left is better) — the hybrid is dominant at high noise. **Right: RMSD to true TS** (median and p95) — the hybrid holds the tightest tail.

Table 2: Wall-time per converged TS (s) and converged-TS RMSD ($\text{\AA}$, p95), $F_{\max} < 0.01$, $n = 287$.

		10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
wall/conv (s)	Hybrid	11.0	12.0	15.7	34.9	65.0	130.7
	Sella	14.5	15.8	23.2	65.2	132.6	393.8
	GAD	47.7	57.1	74.3	156.0	278.5	472.3
p95 RMSD ($\text{\AA}$)	Hybrid	0.047	0.047	0.049	0.055	0.062	0.109
	Sella	0.073	0.071	0.072	0.201	0.617	0.838
	GAD	0.018	0.021	0.028	0.044	0.088	0.456

3.5 Scope boundary: GAD needs to start near the saddle

- Robustness above is for *perturbed-saddle* guesses. From a *reactant* start the GAD family is weak: hybrid $\sim 2\%$, GAD $dt=0.005$ 54% (partial n), while Sella reaches 81% (Fig. 11).
- Interpretation: far from the saddle the reaction mode is not yet identifiable as the lowest Eckart-projected eigenvector, so the GAD reflection has no correct direction to climb. GAD/hybrid are TS *refiners* given a near-saddle guess, not single-ended searchers from a minimum.

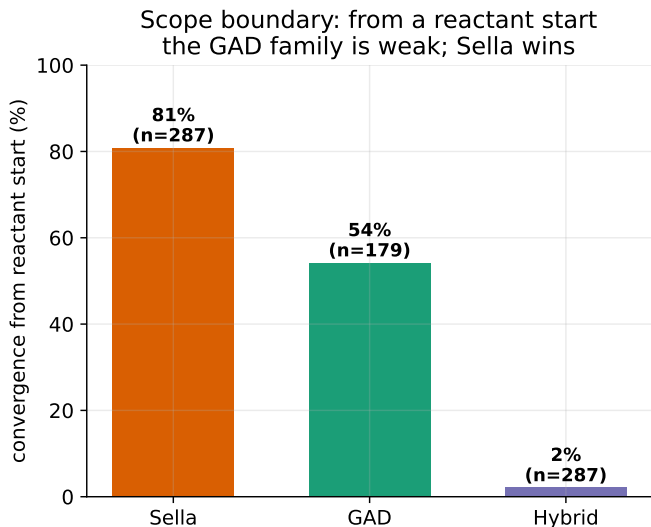


Figure 11: Scope boundary. Convergence from a reactant start: Sella wins; the GAD family needs a near-saddle guess.

4 Discussion

- When the analytic Hessian is cheap, the limiting factor is the step, not the curvature budget. A minimal first-order GAD walk is more robust on the metric that matters (intended success) precisely where large Newton/RFO jumps fail — under a degraded guess — while tying at low noise.
- The failure is *not* wrong basins: across all methods, converged geometries essentially never connect the wrong reaction. The high-noise gap is non-convergence plus one-sided (partial) saddles, which the small in-basin Euler step avoids.

- **Recommended default:** Hybrid GAD–Newton — fastest per converged TS, geometrically tightest, and chemically as reliable as GAD — for near-saddle (refinement) guesses.
- **Limitations:** (i) demonstrated for perturbed-saddle guesses; from a reactant start the GAD family is weak and Sella wins (Section 3.5). (ii) Convergence below $\tau=0.001$ is unreachable for all methods here. (iii) Isotropic perturbations; template/interpolation guesses carry structured error (future work).

Appendix

A Full tables (all nine configurations, $F_{\max} < 0.01$, $n = 287$)

Table 3: Intended success rate % (top) and convergence rate % (bottom), all nine configurations, from the aggregate IRC sweep. Bold = column best. The three headline methods (GAD, Sella cart+Eckart untuned d=1, Hybrid damped) were independently re-validated per-sample (FINDING_IRC_5TIER_2026_06_23; irc_outcomes_5tier_test287.csv); main-text numbers use that fresh re-validation and differ here by ≤ 2 pp, with unintended ≈ 0 confirmed.

Intended %	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD $dt = 0.003$	88.5	89.2	88.9	78.4	61.0	44.6
GAD $dt = 0.005$	88.9	89.2	88.9	78.0	61.7	44.6
GAD $dt = 0.007$	88.5	88.5	88.5	78.4	61.7	43.9
Sella cart tuned d=1	88.9	88.9	87.5	70.7	46.7	17.8
Sella cart+Eck untuned d=1	89.2	89.2	87.5	72.5	49.8	23.3
Sella internal tuned d=1	88.2	87.5	83.3	64.5	33.1	16.0
Sella cart+Eck untuned d=3	88.5	88.9	87.1	70.4	45.3	22.0
Hybrid damped Eckart tr=0.05	89.2	88.9	88.9	76.7	57.5	38.7
Hybrid undamped Eckart tr=0.05	88.9	88.9	88.5	77.0	57.1	38.7
Convergence %						
GAD $dt = 0.003$	89.2	88.5	85.4	71.1	55.1	40.8
GAD $dt = 0.005$	89.2	88.5	85.7	71.8	57.1	43.2
GAD $dt = 0.007$	89.2	88.9	85.7	72.8	58.2	44.6
Sella cart tuned d=1	92.0	91.3	87.5	65.5	42.9	18.8
Sella cart+Eck untuned d=1	92.7	92.0	88.2	70.7	54.0	27.2
Sella internal tuned d=1	79.1	77.4	71.8	50.9	26.8	13.9
Sella cart+Eck untuned d=3	96.5	95.5	92.0	72.8	50.5	23.3
Hybrid damped Eckart tr=0.05	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped Eckart tr=0.05	84.7	84.3	81.5	65.5	49.8	31.0

B Comprehensive $F_{\max}$ sweep (convergence %, $n = 287$)

Table 4: Convergence at five thresholds; GAD block is 0.0% at $F_{\max} < 0.005$ everywhere.

thr	method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
0.05	GAD $dt=0.005$	98.3	97.9	94.8	82.6	71.1	63.1
	Sella $d=1$	98.6	98.3	95.1	83.3	66.2	46.0
	Hybrid damped	97.2	96.9	93.0	78.0	59.6	44.9
0.01	GAD $dt=0.005$	89.2	88.5	85.7	71.8	57.1	43.2
	Sella $d=1$	92.7	92.0	88.2	70.7	54.0	27.2
	Hybrid damped	85.4	85.0	81.5	66.9	50.9	33.1
0.005	GAD $dt=0.005$	0.0	0.0	0.0	0.0	0.0	0.0
	Sella $d=1$	33.8	32.1	32.4	24.7	15.3	7.0
	Hybrid damped	7.3	9.4	10.8	9.1	7.0	6.6

C Convergence-criterion family

Force thresholds (eV/Å): ASE default 0.05; Gaussian “Normal” max-force 0.0231 ($4.50 \times 10^{-4} E_h/a_0$); project canonical 0.01; tight 0.005; Sella documentation 0.001. $F_{\max}$ = maximum force component on any atom.

D Sella baseline configurations

Three axes: {cartesian|internal} {Eckart}? {tuned|untuned} Hess.Freq.= N . Headline baseline: cartesian + Eckart, untuned, Hess.Freq.=1 (exact Hessian every step). Hess.Freq.=3 (library default) raises convergence at low noise (96.5 vs. 92.7% @ 10 pm) but lowers intended success at every noise (Fig. 5) — a stale Hessian yields more one-sided/failed outcomes, not unintended ones.

E Reproducibility

- **Data (committed CSVs):** master_2026_05_16.csv, noise_sweep_with_irc.csv, irc_topo_existing_methods.csv, irc_topo_hybrid.csv, threshold_sweep_2026_05_16.csv, rmsd_to_known_ts_compare.csv, reactant_0pm_2026_05_16.csv, test_summary_full.csv (per-sample).
- **Figures:** make_figs2.py. **Code:** core/gad.py (Eq. 1), projection/projection.py, search/hybrid_gad_eigfollownewton_eckart.py, search/irc_validate.py (Sella IRC + VF2).
- Headline numbers are full $n = 287$. The 200 pm count illustration (130/89/68; 68/101/116) is from the latest local re-scoring and may differ by a few counts from the committed aggregate CSVs; McNemar significance is from the per-sample paired analysis.

References

- [1] W. E and X. Zhou, *Nonlinearity* **24**, 1831 (2011).
- [2] G. Henkelman and H. Jónsson, *J. Chem. Phys.* **111**, 7010 (1999).
- [3] C. J. Cerjan and W. H. Miller, *J. Chem. Phys.* **75**, 2800 (1981).

- [4] A. Banerjee, N. Adams, J. Simons, R. Shepard, *J. Phys. Chem.* **89**, 52 (1985).
- [5] J. Baker, *J. Comput. Chem.* **7**, 385 (1986).
- [6] E. D. Hermes, K. Sargsyan, H. N. Najm, J. Zádor, *J. Chem. Theory Comput.* **15**, 6536 (2019).
- [7] E. D. Hermes, K. Sargsyan, H. N. Najm, J. Zádor, *J. Chem. Theory Comput.* **18**, 6974 (2022).
- [8] M. Schreiner, A. Bhowmik, T. Vegge, J. Busk, O. Winther, *Sci. Data* **9**, 779 (2022).
- [9] A. Burger, L. Thiede, N. Rønne, *et al.*, “Shoot from the HIP: Hessian interatomic potentials without derivatives,” arXiv:2509.21624 (2025).
- [10] E. C.-Y. Yuan, *et al.*, *Nat. Commun.* **15**, 8865 (2024); arXiv:2405.02247.
- [11] G. Wu, C.-Y. Yuan, K. Hegazy, S. M. Blau, T. Head-Gordon, arXiv:2603.21323 (2026).